SCIENCE HISTORY INSTITUTE

RONALD D. MACFARLANE

Transcript of an Interview
Conducted by

Michael A. Grayson

at

Texas A&M University
College Station, Texas

on

26 May 2011

(With Subsequent Corrections and Additions)

Ronald D. Macfarlane

ACKNOWLEDGMENT

This oral history is one in a series initiated by the Chemical Heritage Foundation on behalf of the American Society for Mass Spectrometry. The series documents the personal perspectives of individuals related to the advancement of mass spectrometric instrumentation, and records the human dimensions of the growth of mass spectrometry in academic, industrial, and governmental laboratories during the twentieth century.

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RONALD D. MACFARLANE

1933	Born in Buffalo, New York, on 21 February

Education

1954	BA, University of Buffalo, Chemistry
1957	MS, Carnegie Institute of Technology, Chemistry
1959	PhD, Carnegie Institute of Technology, Chemistry

Professional Experience

	University of California, Lawrence Berkeley Laboratory
1959-1962	Postdoctoral Fellow
	McMaster University
1962-1965	Assistant Professor of Chemistry
1965-1967	Associate Professor of Chemistry
	Texas A&M University
1967-2008	Professor of Chemistry
2008-present	University Distinguished Professor

Honors

1969	J. Simon Guggenheim Fellow, Niels Bohr Institute, Copenhagen
1974	Visiting Faculty, Center for Nuclear Research, Strasbourg, France
1974	Visiting Faculty, Phillips University, Marburg, Germany
1981	Visiting Faculty, University of Uppsala, Sweden
1982	University of Paris, Orsay, France
1984	Texas A&M Faculty Distinguished Achievement Award for Research
1989	American Chemical Society Award in Nuclear Chemistry
1991	American Society for Mass Spectrometry Distinguished Achievement Award (Inaugural Award)

ABSTRACT

Ronald Macfarlane was born in Buffalo, New York, the oldest of three children. An excellent teacher in high school sparked his interest in chemistry, and Macfarlane attended the University of Buffalo, majoring in analytical chemistry. He found coursework rather boring, but relished exciting summer jobs in chemical industries. Nuclear chemistry was just getting started, and Macfarlane entered Carnegie Institute of Technology for a PhD. In Truman Kohman's lab, he researched natural radioactivity. He made a kind of giant Geiger counter, which he published to international praise. Next, he accepted a postdoctoral position at Lawrence Berkeley National Laboratory, working on alpha activity in rare earth elements. After accidentally creating a more efficient way to get ionized particles; he discovered new isotopes for years, saving his discoveries for later publication.

Macfarlane accepted a job at McMaster University. There, he named his accidental creation the "helium jet recoil method" and began publishing data he'd stored up. He visited the Soviet Union, where he met John McIntyre, a physics professor at Texas A&M University. Months later, Arthur Martell, chairman of the new chemistry department at Texas A&M, called to recruit Macfarlane, and he took up a full professorship there.

The Atomic Energy Commission funded Macfarlane's nuclear work for a while but ceased after an incidental discovery during one of his nuclear chemistry experiments led to what became known as ^{252}californium plasma desorption mass spectrometry. Macfarlane left the nuclear chemistry field to concentrate on mass spectrometry. He spent fifteen years developing the method that was the first to characterize the mass of large, fragile biomolecules—a method that quickly became well known and widely used to characterize a wide spectrum of biomolecules especially in the pharmacy and medicine fields. Early in the course of the discovery, he obtained National Institutes of Health funding to develop and expand the methodology. As one discovery led to another, his focus drilled down to another new field involving characterization of unusual lipids. He believed in "letting nature tell [a person] what is going on;" this approach has led to his interest in trying to determine who has cardiovascular disease and which components of his or her lipid profile contribute to the disease. One of the most important discoveries involved the characterization of an atherogenic type of the good cholesterol associated with APOC-1 (apolipoprotein C1), using both mass spectrometry and some of the novel platforms his lab developed to characterize lipoproteins.

At the time of this interview, Macfarlane, age seventy-eight, was still unready to retire. Having thrown out the textbook in favor of his own "commentaries," he continued to teach analytical chemistry his way, incorporating constructivism, conceptual learning, and other elements of educational psychology. Using blood samples from actual patients Macfarlane continued his work on cardiovascular disease. He believes that a person should contribute to the betterment of society, which he thinks he has done. His work, which has received nearly continuous funding, has straddled the boundary between applied and pure science, and he has always wished he could return to "real science." Macfarlane concludes the interview by saying that his colleagues over the years have been supportive and gracious; most of his collaborations have worked equitably; he has tried to mentor his students while fostering their own creativity. Macfarlane's advice to young scientists is to listen to nature and to pay attention to small details.

INTERVIEWER

Michael A. Grayson is a member of the Mass Spectrometry Research Resource at Washington University in St. Louis. He received his BS degree in physics from St. Louis University in 1963 and his MS in physics from the University of Missouri at Rolla in 1965. He is the author of over 45 papers in the scientific literature. Before joining the Research Resource, he was a staff scientist at McDonnell Douglas Research Laboratory. While completing his undergraduate and graduate education, he worked at Monsanto Company in St. Louis, where he learned the art and science of mass spectrometry. Grayson is a member of the American Society for Mass Spectrometry (ASMS), and has served many different positions within that organization. He has served on the Board of Trustees of CHF and is currently a member of CHF's Heritage Council. He currently pursues his interest in the history of mass spectrometry by recording oral histories, assisting in the collection of papers, and researching the early history of the field.

ABOUT THIS TRANSCRIPT

This interview was conducted as part of the Mass Spectrometry Oral History project, a collaboration between the Science History Institute and the American Society for Mass Spectrometry. The Mass Spectrometry Oral History project records the human dimensions of the growth of mass spectrometry in academic, industrial, and governmental laboratories during the twentieth century.

The Center for Oral History, Science History Institute, is committed both to preserving the recording of each oral history interview in our collection and to enhancing research use of the interviews by preparing carefully edited transcripts of those recordings. The preparation of interview transcripts begins with the creation of a verbatim typescript of the recording and proceeds through review and editing by staff of the Center; interviewees also review the typescript and can request additions, deletions, or that sections be sealed for specified periods of time. We have established guidelines to help us maintain fidelity to the language and meaning of each recorded interview while making minor editorial adjustments for clarity and readability. Wherever possible, we supply the full names of people, organizations, or geographical locations mentioned during the interview. We add footnotes to the transcript to provide full citations for any publications that are discussed, to point to extant oral history interviews, and to clear up misstatements or provide context for ambiguous references in the transcript. We use brackets to indicate the addition of material that was not in the audio, and bracketed ellipses to indicate the deletion of recorded material. The transcript also includes time stamps at five-minute intervals. We omit without noting most instances of verbal crutches and all instances of nonlexical utterances. We also make small grammatical corrections where necessary to communicate interview participants' meaning. Finally, staff of the Center create the abstract, chronology, table of contents and index.

Audio quality, particularly of the interviewee as opposed to the interviewer, is noticeably poor overall, and some portions are inaudible.

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INTERVIEWER: **Michael A. Grayson**

LOCATION: **Texas A&M University, College Station, Texas**

DATE: **26 May 2011**

GRAYSON: So, I'm going to start the way we usually do by saying, my name is Mike Grayson, and I'm sitting in the office of Professor Ron Macfarlane in Texas A&M University in College Station, Texas. Today is 26 May 2011. A nice pleasant day, but it'll get warm before everything's over. So, we're going to do a little oral history interview of Ron's career. Normally, we like to start these back in the beginning of time, so do we have information on your birthdate and location? I don't think so.

MACFARLANE: No.

GRAYSON: So when were you born?

MACFARLANE: Nineteen thirty-three, 21 February, in Buffalo, New York.

GRAYSON: Okay. So you're a real transplant to this part of the country.

MACFARLANE: That's right.

GRAYSON: And your parents' names?

MACFARLANE: William Arthur MacFarlane and Margaret Ann Baker.

GRAYSON: Okay. Were there any other children in the family, or you were . . . ?

MACFARLANE: Yes. I was the oldest. I have a brother and a sister.

GRAYSON: In the beginning you obviously had developed an interest in science somewhere along the way . . .

MACFARLANE: [Yes], right.

GRAYSON: So, a little bit about your education—well, how about your parents' education? Were they very well educated?

MACFARLANE: No, they weren't. They came through the Depression.

GRAYSON: Ah, okay.

MACFARLANE: And so neither one of them had a college education. In fact, I think my dad went to high school. I'm not for sure about my mom. My mother was born in Virginia—Richmond, Virginia. My father was Canadian.

GRAYSON: Oh, okay.

MACFARLANE: He came down to Buffalo to look for work during the Depression. That's where they met, in Buffalo.

GRAYSON: I see.

MACFARLANE: They settled there, and that's where the family started. Then I went to a public school there, and then I went to high school—Kensington High School.

GRAYSON: Is that Kensington, did you say?

MACFARLANE: Kensington High School, [yes]. And in retrospect—I'll just put in some things that, in retrospect, had an influence on the direction of my career.

GRAYSON: What was that first one again?

MACFARLANE: Gadolinium. So he said, "Why don't you look at that?" I said okay. What you do is you take a small sample—a few micrograms of material. You put it on a photographic plate. You let it sit for a couple of months. Then after that, you develop it, and you see tracks through the alpha particles. You count the tracks. After a couple of months of that I thought, "This sucks." I said to Dr. Kohman, I said, "There's got to be a better way to do this." "What do you have in mind?" I thought about what the problem is: "There's not enough material there. I need to have a detector that I can put more material in." "What do you have in mind?" I said, "Well, a giant Geiger counter." I said, "Something this big." Here again, it was the daydreaming part. So to my amazement, he said, "Go for it."

GRAYSON: Oh, wow, okay.

MACFARLANE: "I'll fund it and you go into it." So I spent the next three years putting this together.

GRAYSON: So this was an improvement to the existing technology—

MACFARLANE: That's right—

GRAYSON: —for your particular problem. So I <**T: 15 min**> guess the disintegrations that were occurring were at such a slow rate—

MACFARLANE: That's right. The half-lives were like ten to fifteen years.

GRAYSON: Oh, okay.

MACFARLANE: So, I got the darn thing working, and I remember he had an electrical engineer in his group. He said, "I'd be willing to help you." I said, "I don't want any help." It was at a time where you didn't have electronics, you had to build your own electronics. So I did that. I learned how to do that. I built my own electronics. This engineer did it. He developed what's called pulse-height analyzer. I used that. I showed Dr. Kohman that the thing really works. He was very excited about that. He said, "Well, go for it." My thesis then was on that,

and I made some discoveries. So I got a thesis out of that. It was published in *Physical Review*.[1] He got all sorts of letters of congratulations from colleagues around the world. I never thought about science being international. I thought this is really cool. He showed it to me. So then I realized the instrument that I had, nobody else has. I had the whole field to myself.

GRAYSON: You had developed something that was unique.

MACFARLANE: [Yes].

GRAYSON: How did you [. . .] get the super Geiger counter? I mean, what's the secret to it?

MACFARLANE: Well, first of all, it was a big cylinder. It was about that long and about that diameter. I had sheets of copper that I could—I had a way of actually putting a solution of the sample on a trough. Then I would slowly rotate the sheet, while there was a heat lamp on it. So that way, I got a uniform distribution of that on the cylinder. In the literature there was a different kind of Geiger counter called a proportional counter—

GRAYSON: The what?

MACFARLANE: Proportional counter—

GRAYSON: Oh, proportional counter.

MACFARLANE: —which consisted of a cylinder with a wire going down the middle. So then in a vacuum an alpha particle—not a total vacuum. The alpha particle goes out, it ionizes the gas in the proportional counter, and the electrons are attracted to this wire, which is positively charged. [fingers snapping] It produces an electronic pulse. So I count them. Also the magnitude of the pulse is proportionate to the energy of the alpha particles. So I was able to use the spectrum—energy spectrum—of the alpha particles.

GRAYSON: And so this is just a really basic, scaled-up concept of a different, smaller version of—

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[1] R. D. Macfarlane. "Natural Alpha Radioactivity in Medium-Heavy Elements," *Physical Review* 121 (1961): 1758-1769.

MACFARLANE: [Yes]. Right

GRAYSON: —that had been created by somebody. But it was good for you.

MACFARLANE: Right. But it not only counts the particles, it tells you what the energy of the alpha particles is. And so there was one isotope of samarium that was well known—samarium-147. I put that on there, and they just lit up. It was unbelievable. I counted the sensitivity of the [. . .] at that time.

I was also a TA. I got an exposure to actually doing some teaching. I realized this is something that it's a possibility that I want to stay in academia, and because I was reasonably good at it, but I need more time. So then I told Dr. Kohman, "I think I'd like to take a postdoctoral appointment. I need a little bit more time to figure out what I want to do."

GRAYSON: Well, did you tell your mother that too? [laughter]

MACFARLANE: Well, at this point, she was very happy, and particularly when I got my Ph.D. She was on a cloud. She sent an article to the Buffalo newspaper and a picture and all sorts of stuff. I didn't know this, but after she died I found a scrapbook with all of this in it.

GRAYSON: Oh, wow, okay.

MACFARLANE: I asked Dr. Kohman. I said, "Where should I go?" He said, "Well, the best place is [University of California] Berkeley. Lawrence [Berkeley National Laboratory] Radiation Laboratory. A good friend of mine is Glen [T.] Seaborg."

GRAYSON: Oh, wow.

MACFARLANE: He was a Nobel Laureate [Chemistry, 1951] and all that stuff. So I applied, and he sent a support letter, and I got accepted. So then I arrived at Lawrence Berkeley Laboratory, that's the mecca for nuclear chemistry. I arrived and I said to Dr. Seaborg, "Well, what do you want me to do?" <**T: 20 min**> He says, "Anything you want to. This is a lab . . . whatever you want to do." I did not anticipate this. [laughter]

MACFARLANE: Well, [yes]. I realize I'm lucky. And but here again, this is like what Frank [Field] said that I use my nose to lead the way.[28] That's what's <**T: 90 min**> made it so exciting for me, the way my career went. It all goes back to my graduate days with Dr. Kohman and doing things that are out of the box. And when you're in the box, you're a little too sheltered.

GRAYSON: [Yes], and, you know, there's other people in the box with you. [laughter]

MACFARLANE: [Yes], that's right. That's right, [yes].

GRAYSON: Okay. Well, I'll see if you have any other things you want to say at this particular point in time. I think we can say we pretty well covered the turf. Invariably something will come to mind after the fact.

[END OF AUDIO, FILE #1.2]

[END OF INTERVIEW]

[28] Frank Field, Oral History Transcript #0352.

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